Chipsee

Industrial HMI Display

EPC-ESP32P4-070



PN: CS10600-ESP32P4-070E

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EPC-ESP32P4-070



Front View



Rear View



Side View 1



Side View 2

1 Product Overview

The EPC-ESP32P4-070 Embedded HMI Display (PN: CS10600-ESP32P4-070E) is an embedded programmable capacitive touch HMI display featuring an ESP32 P4 processor. The product features an embedded mounting design, making it suitable for integration into existing equipment.

It features a 7” IPS display with a maximum brightness of 400 cd/m^2. The touch screen is a responsive capacitive screen and supports multi-touch.

The product is powered by an ESP32-P4 processor, making it suitable for applications such as:

- Smart Home
- Industrial Automation
- Healthcare
- Consumer Electronics
- Smart Agriculture
- Retail Self-Service Terminals (POS, Vending Machines)
- Service Robots
- Multimedia Player
- High-Speed USB Host and Device

- Smart Voice Interaction Terminal
- Edge Vision AI Processor
- HMI Control Panel

The product can be programmed with ESP-IDF, ESPHome and other ESP32 ecosystem tools and frameworks that support the ESP32-P4.

2 Ordering Options

Chipsee products can be customized during the ordering process. The product will be shipped with the pre-installed factory defaults if no extra requirements are specified. The table in the Hardware Features section provides information about the default options bundled with the product.

> **Note**
>
> You can order EPC-ESP32P4-070 from the official Chipsee Store or from your nearest distributor.

3 Hardware Features

The EPC-ESP32P4-070 Embedded HMI Display offers a broad range of performance and connectivity features for flexible integration. Some of the key features are listed below.

Key Features

EPC-ESP32P4-070	
CPU	Dual-core 32-bit RISC-V, up to 400 MHz; single-core 32-bit LP RISC-V, up to 40 MHz
Storage	32MB external SPI Flash
RAM	768 KB HP L2MEM + 32 KB LP SRAM + 8 KB SPM; 32MB in-package PSRAM
FRAM	Optional, not mounted by default
Gyroscope	Optional, not mounted by default
Display	7” IPS LCD, 1024 x 600 px, brightness 400 cd/m^2
Touch	5-point capacitive touch with 1.0mm tempered glass
LAN	1 x 100Mbps LAN
PoE	Power over Ethernet 802.3af/at, optional, not mounted by default
WiFi/BT	WiFi 6, Bluetooth LE 5 **(optional, ESP32-C6 is not mounted by default)**
4G/LTE	Internal Cat 1 4G/LTE module, not mounted by default
Audio	3.5mm audio in/out interface; 2W internal speaker
Buzzer	Onboard buzzer, driven by I2C expanded IO
GPIO	8 channels isolated I/O, 4 x input and 4 x output. 4 x direct ESP32-P4 GPIO (non-isolated, 3.3V)
RS232	2 x RS232 **(default)**; up to 4 x RS232 **(optional, by configuring RS485 to RS232)**
RS485	2 x RS485 **(default)**; 2 x RS485 can be configured to 2 x RS232 **(optional)**
CAN	1 x CAN **(default)**

EPC-ESP32P4-070	
USB	2 x USB 2.0 Type-A high-speed host ports; 1 x USB 1.1 Type-C port for firmware flashing and debugging; the type-C port supports powering the device while programming
Micro SD card socket	1 x TF (micro SD) card slot
RTC	High-accuracy RTC with supercapacitor, can work 1 week after power off **(default)**. High accuracy RTC with lithium coin battery, can work 3 years after power off **(optional)**.
Power Input	9V to 32V DC
Current	260mA max at 15V, 200mA typical at 15V
Power Consumption	4W max, 3W typical
Operating Temp.	From -10°C to +60°C
Dimensions	190 x 107.8 x 27.7mm
Weight	400g
Mounting	Embedded
Certification	CE, RoHS

3.1 Optional Features

- WiFi/Bluetooth: If Wi-Fi/Bluetooth connectivity is required, an ESP32-C6 module can be installed at the factory. The ESP32-C6 is not populated by default.
- 4G/LTE: 4G/LTE (Cat 1) module is supported on this product, it's not mounted by default, if you need the interface or the 4G/LTE module you can contact us when you place an order.
- Gyroscope is supported but not mounted by default, if you need gyroscope, contact us before placing an order.
- FRAM (Ferroelectric Random Access Memory) is supported but not mounted by default, if you need FRAM, contact us before placing an order.
- 15 pin CSI camera interface is customizable, if you want to use a camera (like Raspberry Pi camera V1.3 OV5647) on the product, contact us.

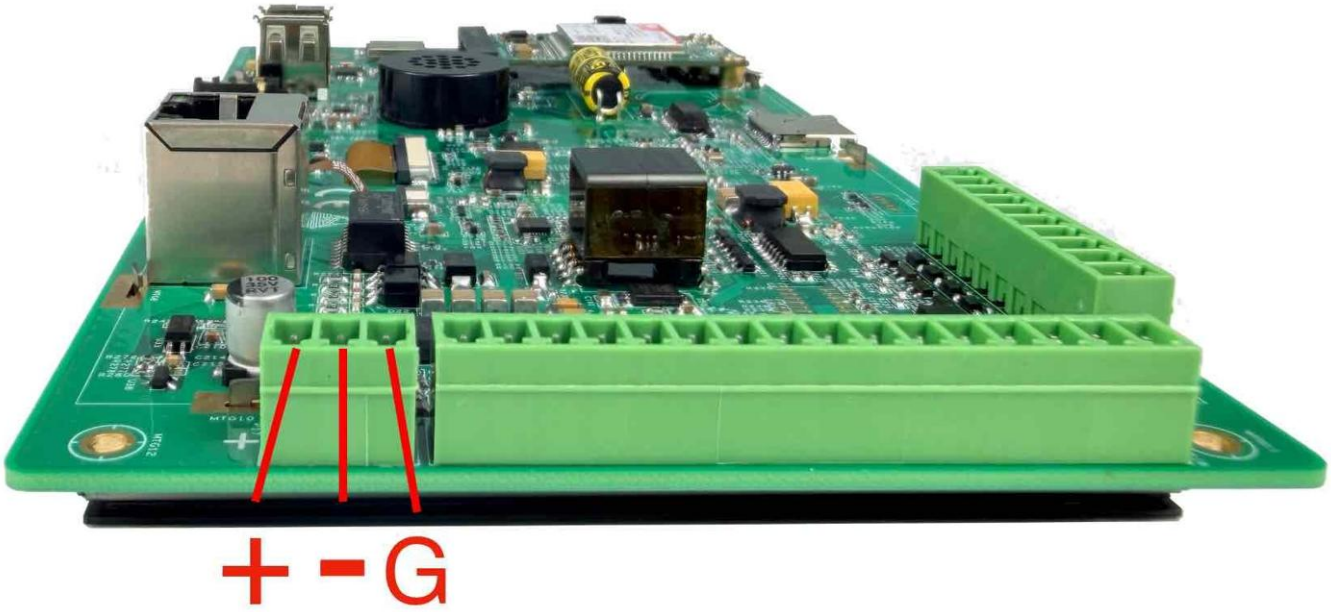
> ⚠ **Warning**
>
> Installation, repair, and maintenance tasks should be performed by trained personnel only. Chipsee does not bear any responsibility for damage caused by inadequate handling of the product.

4 Power Input

The EPC-ESP32P4-070 Embedded HMI Display can be powered by an input voltage of 9V to 32V DC.

The total power consumption is about 4W max, 3W typical, depending on the load and brightness.

The power input is **3-pin 3.81mm screw terminal**, you can also use the USB-C port to power the product while programming.



Power Connector



USB Power Input

Note that the "**+**" sign represents the positive power input, the "**-**" terminal is shorted to the ground.

Power Connector

Power Input Definition		
Pin Number	**Definition**	**Description**
Pin 1	Positive Input	DC Power **Positive Terminal**
Pin 2	Negative Input	DC Power **Negative Terminal**
Pin 3	Ground	**Power System Ground**

> **Note**
>
> The system ground "**G**" is connected to power negative "**-**" on board.
>
> The product supports **optional** PoE (Power over Ethernet), it's **not** mounted by default.

5 Connectivity

There are many connectivity options available on the EPC-ESP32P4-070.

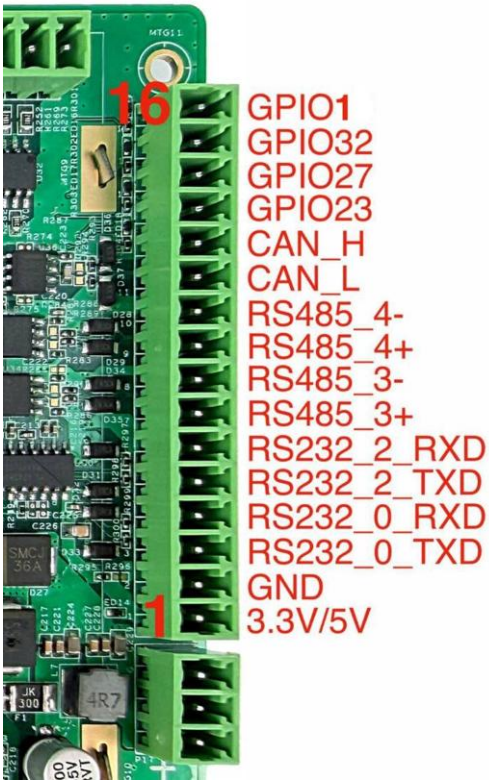
5.1 RS232/RS485/CAN/Direct GPIO

The product provides the following serial and CAN interfaces:

- 2 x RS232 **(default)**; up to 4 x RS232 (**optional**, by configuring RS485 to RS232)
- 2 x RS485 **(default)**; 2 x RS485 can be configured to 2 x RS232 **(optional)**
- 1 x CAN **(default)**
- 4 x direct ESP32-P4 GPIO

RS232/RS485/CAN Direct GPIO Connector

RS232 / RS485 / CAN / Direct GPIO Pin Definition			
Pin Number	**Definition**	**Description**	**ESP32 P4 Pin**
Pin 16	GPIO1	Direct ESP32-P4 GPIO	GPIO1
Pin 15	GPIO32	Direct ESP32-P4 GPIO	GPIO32
Pin 14	GPIO27	Direct ESP32-P4 GPIO	GPIO27
Pin 13	GPIO23	Direct ESP32-P4 GPIO	GPIO23
Pin 12	CAN_H	CAN high signal	GPIO22
Pin 11	CAN_L	CAN low signal	GPIO21
Pin 10	RS485_4-	RS485 –(B) signal	GPIO47
Pin 9	RS485_4+	RS485 +(A) signal	GPIO48
Pin 8	RS485_3-	RS485 –(B) signal	GPIO4
Pin 7	RS485_3+	RS485 +(A) signal	GPIO5
Pin 6	RS232_2_RXD	RS232 RXD signal	GPIO2
Pin 5	RS232_2_TXD	RS232 TXD signal	GPIO3
Pin 4	RS232_0_RXD	RS232 RXD signal	GPIO38
Pin 3	RS232_0_TXD	RS232 TXD signal	GPIO37
Pin 2	GND	System ground	GND
Pin 1	3.3V/5V	Power output	3.3V/5V



RS232/RS485/CAN/Direct GPIO Connector

Attention

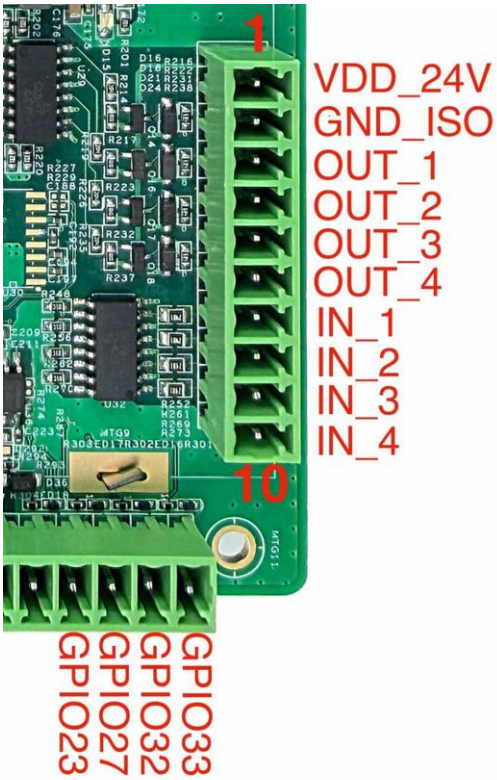
1. Pins 13 to 16 are direct ESP32-P4 GPIO signals. They are 3.3V logic and not 5V tolerant.
2. Pin 1 is 5V by default, if you need 3.3V, you can **Contact us** before shipping.
3. RS485_3, RS485_4 automatically control input/output direction. They do not require software control.
4. The 120 Ω termination resistor for the RS485 is **not mounted** by default.
5. If you need a different configuration, you can **Contact us** before shipping, we can make a customized version of these pins for you.

5.2 Isolated GPIO

The EPC-ESP32P4-070 Embedded HMI Display has 8 optically isolated I/O channels, as shown in the figure below. These are controlled through an I2C GPIO expander at address 0x21.

GPIO Connector Pin-out

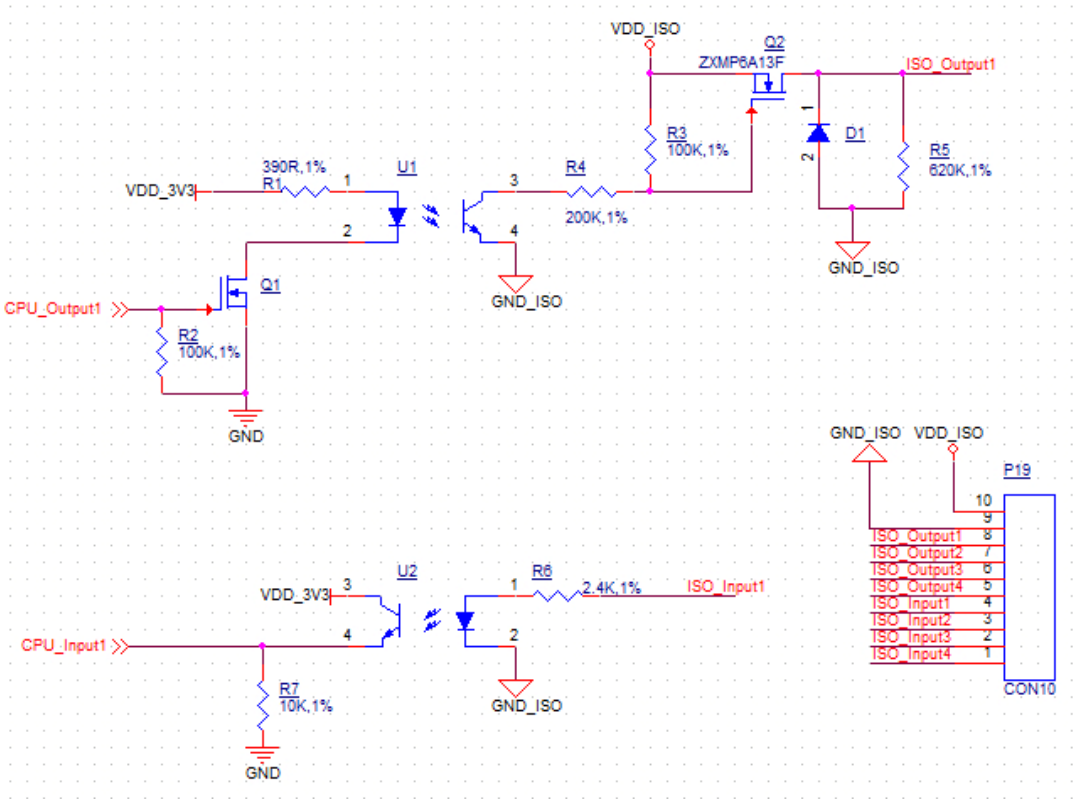
Isolated GPIO Connector Pin Definition		
Pin Number	**Definition**	**Description**
Pin 1	VDD_24V (VDD_ISO)	Isolated Power +5V ~ +24V Input (Should not exceed 24V)
Pin 2	GND_ISO	Isolated Ground
Pin 3	OUT1	Isolated Output 1
Pin 4	OUT2	Isolated Output 2
Pin 5	OUT3	Isolated Output 3
Pin 6	OUT4	Isolated Output 4
Pin 7	IN1	Isolated Input 1
Pin 8	IN2	Isolated Input 2
Pin 9	IN3	Isolated Input 3
Pin 10	IN4	Isolated Input 4



Optical Isolated GPIO Connector

Attention

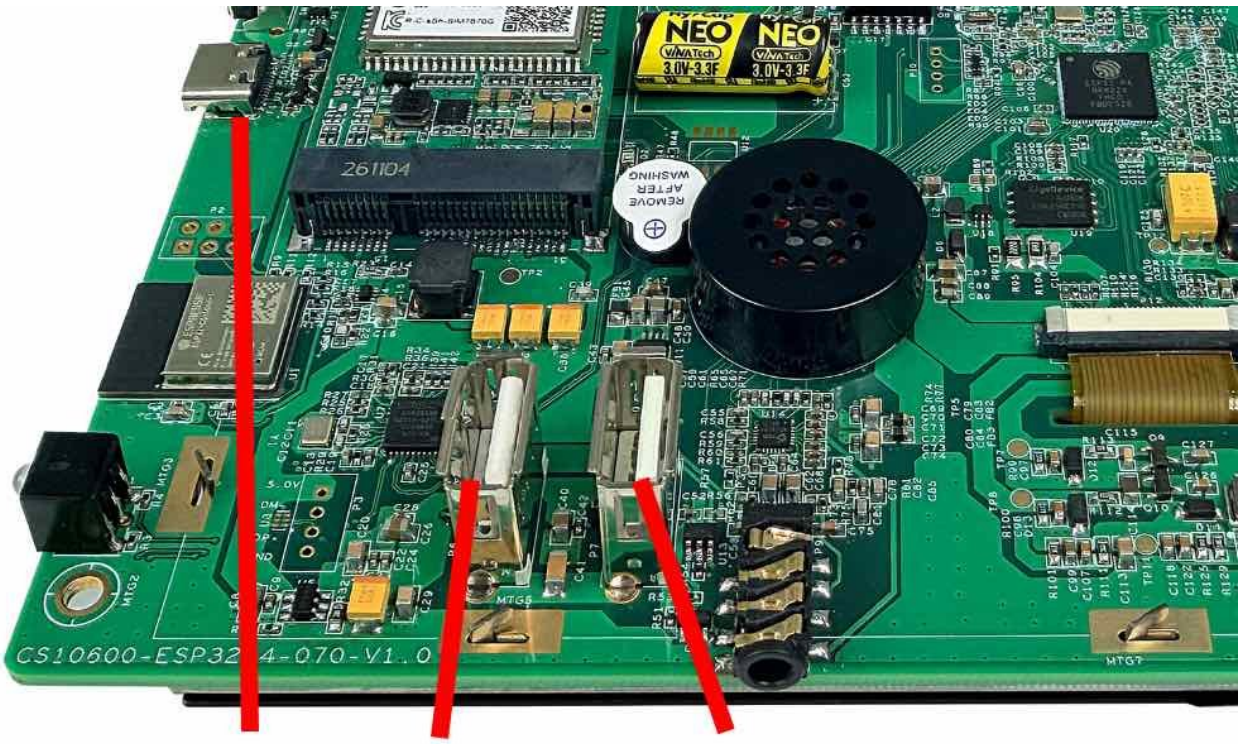
1. In order to use the Isolated Output, you need to add an external Isolated Power to the VDD_ISO and GND_ISO (between 5V and 24V). The power voltage should not exceed 24V.
2. The output current can reach 500mA for every channel, but it also depends on the isolated source that supplies the power.
3. In order to use the Isolated Input, you need to add a signal to the IN_X and GND_ISO. A 2.4KΩ resistor, as R6, has been added to limit the input current, as shown in the figure below. This resistor should work well for the 5-24V input signal. If your input signal is less than 5V, please change this input resistor. The reduced schematic is for reference purposes, if you need the precise resistor schematic, please contact us.



Isolated GPIO reduced schematic

5.3 USB

There are 2 x **USB2.0 Type-A HOST connectors** and 1 x **Type-C connector** onboard, as shown in the image below.



USB2.0 Host and Type-C Ports

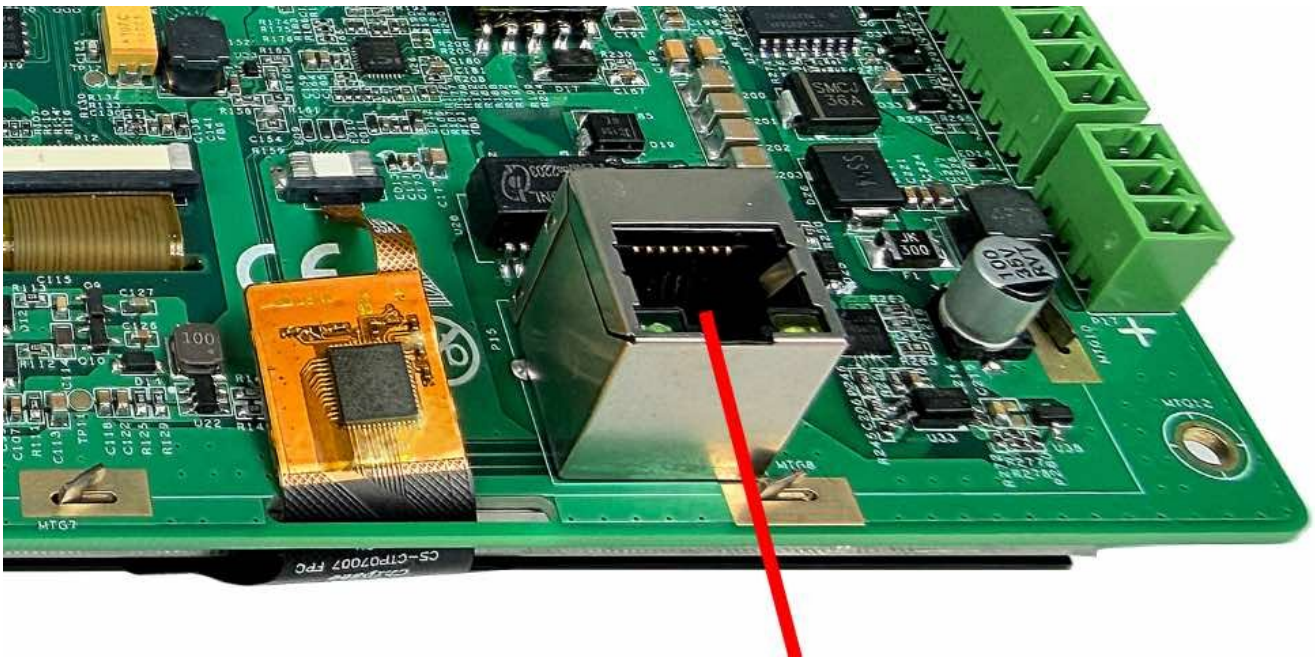
Note

- USB peripherals such as high-speed flash drives can be connected to the USB 2.0 host ports.
- Full/Low speed USB devices such as mice and keyboards **cannot** connect due to **the limitation of USB Hub Host software stack**, Espressif managed software stack doesn't support Transaction Translator layer yet.
- The USB Type-C port is used for debugging and firmware flashing and supports powering the product.

5.4 LAN

LAN (RJ45) connector provides Ethernet connectivity over standardized Ethernet cable. The integrated Ethernet interface supports up to 100 Mbps data throughput.

The product supports **optional** PoE (Power over Ethernet), it's **not** mounted by default.



RJ45 LAN Connector

5.5 Wi-Fi and Bluetooth

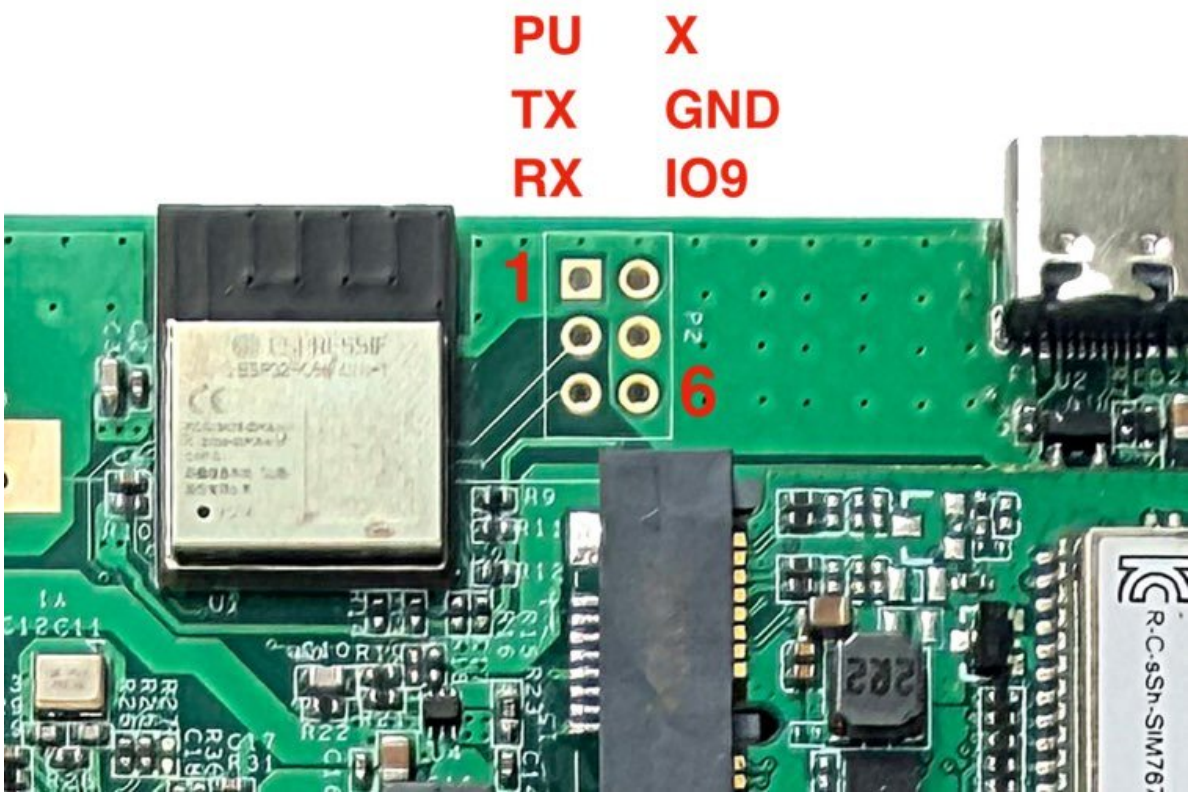
When the optional Wi-Fi/Bluetooth configuration is selected, an ESP32-C6 module provides Wi-Fi and Bluetooth connectivity to the ESP32-P4 through SDIO.

The ESP32-C6 operates as a coprocessor.

Espressif's official ESP-Hosted Wi-Fi and Bluetooth firmware will be flashed at the factory before shipping.

ESP32-C6 Programming Connector Pin Definition

Pin Number	Definition	Description
Pin 1	C6_CHIP_PU	ESP32-C6 chip enable/reset signal
Pin 2	X	
Pin 3	C6_U0TXD	ESP32-C6 UART0 transmit signal
Pin 4	GND	System ground
Pin 5	C6_U0RXD	ESP32-C6 UART0 receive signal
Pin 6	C6_IO9	ESP32-C6 boot mode selection signal



ESP32-C6 Pin Definition

You can flash the C6 firmware by yourself too. The process is the same as the official Espressif ESP32C6 firmware flashing guide.

The ESP32-C6 firmware can be updated using either of the following methods:

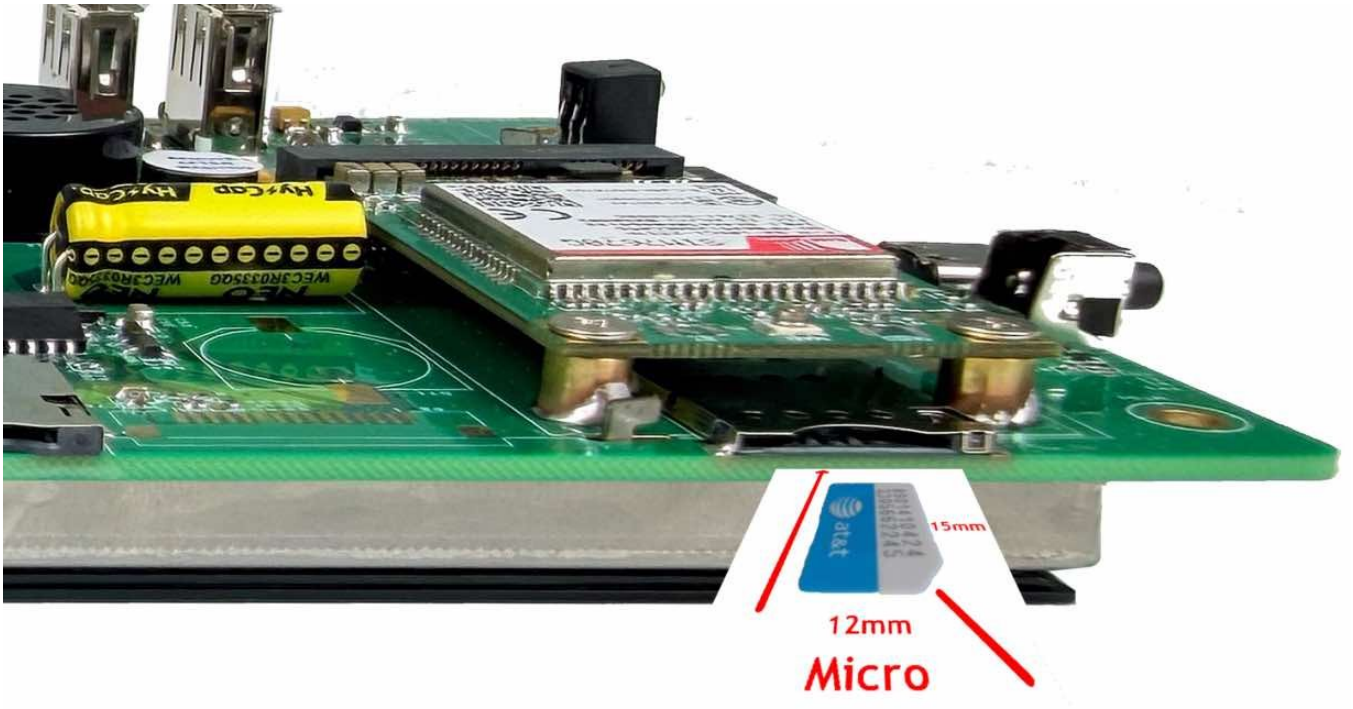
- **UART flashing:**
  1. Prepare a USB-UART (TTL level) cable, to connect the pins and your computer.
  2. Connect GND (Pin 4) to your UART cable's GND, connect GND to C6_IO9 (Pin 6) to let C6 enter flash mode.
  3. Connect C6_U0TXD (Pin 3) to your cable's RX pin.
  4. Connect C6_U0RXD (Pin 5) to your cable's TX pin.
  5. Reset the product/C6 to enter download mode, you can unplug all power sources first and then connect USB-C to your power adapter as a reset.
  6. Remove the C6_IO9 (Pin 6) short to leave flash mode in the next reset.
  7. Reset the product/C6 after flashing to boot normally.
- **Updates through the ESP32-P4:** The ESP32-P4 can flash new firmware to the ESP32-C6 through SDIO using a firmware image if C6 already has a firmware (OTA). This is usually done with a MicroSD card.

5.6 4G/LTE

4G/LTE Cat 1 module is **optional**. If you place an order with 4G module included, we will also add a SIM card holder and a 4G/LTE antenna to enable 4G/LTE connectivity.

The SIM card slot does **NOT** support hot-swapping. **Power off** the device before inserting or removing the SIM card.

The Cat 1 module doesn't support GPS.



SIM Card Direction

> **Attention**
>
> The product does not come shipped with the 4G/LTE module by default. If you need to use 4G/LTE, you can contact us when placing an order. We can install the necessary hardware for you.

5.7 MicroSD Card

The EPC-ESP32P4-070 Embedded HMI Display features 1 x **TF Card (micro SD) slot** (note the direction of the micro SD card). The product does not come shipped with the micro SD card by default.

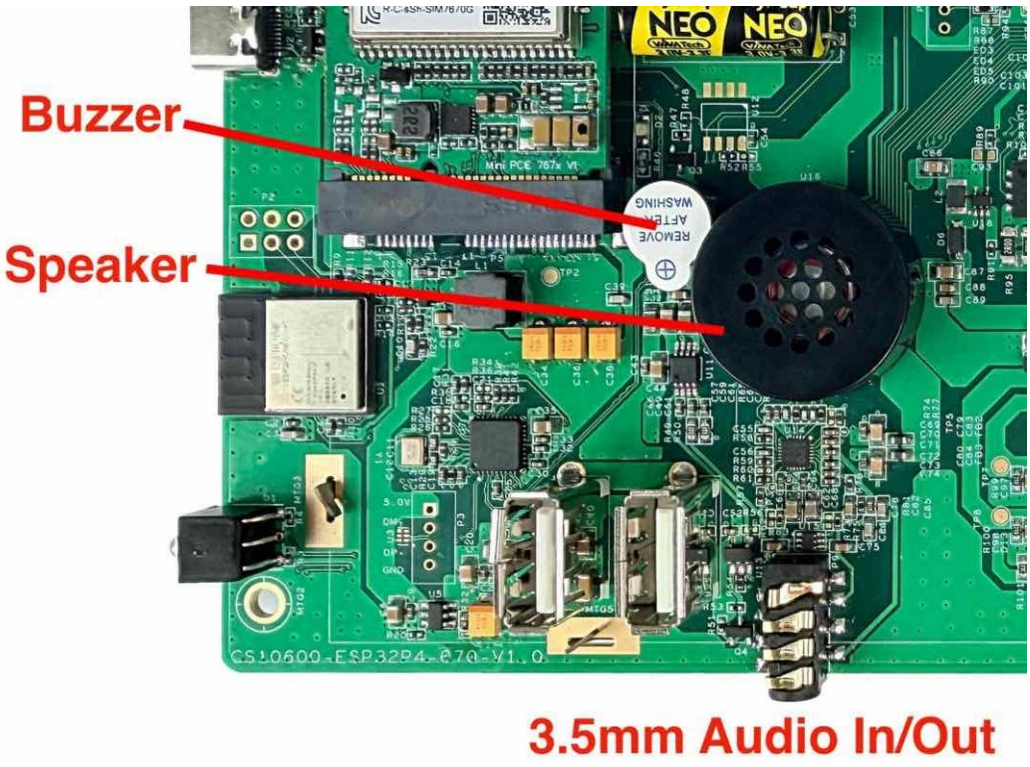


Micro SD Card Slot

5.8 Audio

5.8.1 3.5 mm Audio In/Out

The product features a 3.5mm audio connector as shown in the image below.



Audio Connector

This connector supports both earphone output and microphone input.

There is no on board microphone for this product.

After inserting a 3.5mm device, the on board speaker will be muted.

5.8.2 Internal Speaker

The product also has an internal 2W speaker, meaning you can play music without an earphone, the sound can be played from the product directly.

5.8.3 Buzzer

The product has an internal buzzer, suitable for playing beep sounds for alarms.

6 Button and Status LED

6.1 Status LED

This product has a bi-color status LED. It is green whenever the product is powered on. The application can switch it to red through pin P16 of the on-board PCA9535 I/O expander (I2C address 0x21, on the I2C bus GPIO7 SDA / GPIO8 SCL): set P16 as an output and drive it low for red, high (or leave it as an input) for green only.

P16 is bit 6 of port 1: configuration register 0x07, output register 0x03.

The I/O expander also controls other on-board functions (touch reset, speaker enable, buzzer, isolated I/O). When changing P16, read the current register value first and change only bit 6.



Status LED

6.2 Button

There is a PROG button for flashing firmware.



Button

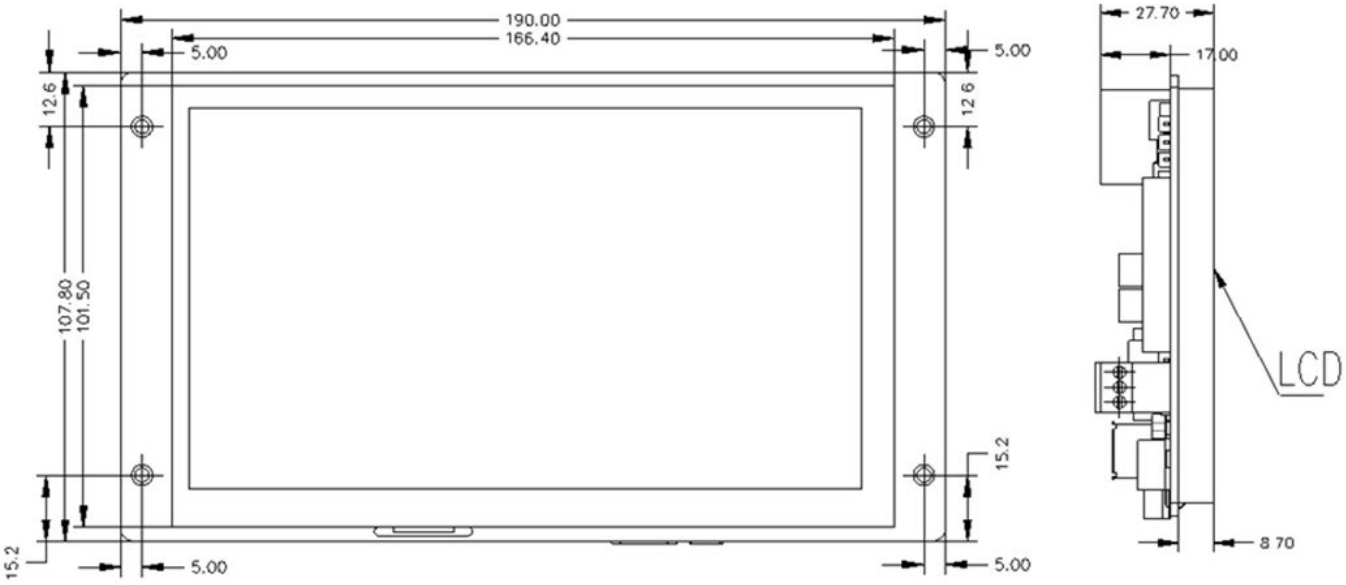
7 Mounting Procedure

The EPC-ESP32P4-070 can be installed in an embedded enclosure (guide).

> **Attention**
>
> Do not apply excessive mechanical pressure to the display when mounting it in an enclosure.

8 Mechanical Specifications

The outer mechanical dimensions of EPC-ESP32P4-070 are 190 x 107.8 x 27.7mm (W x H x D). Please refer to the technical drawing in the figures below for details related to the specific product measurements.



Mechanical Drawing

9 Disclaimer

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